

Title: Ranking Signals in Search Visibility: An Empirical Analysis of CTR, Impressions, and Position

Abstract

This study investigates which ranking signals most strongly correlate with search visibility and engagement. Using FlyRank’s internship dataset, we analyze CTR, impressions, and position as predictors of click behavior. A baseline logistic regression is compared against tree-based models to evaluate predictive power. Results show impressions and position are dominant drivers, while CTR highlights underperforming opportunities. Findings support actionable recommendations for metadata optimization and monitoring. Limitations include observational scope and non-causal inference, but the framework provides decision-support for ranking improvements.

Introduction / Problem Statement

Search teams often ask: *Which signals matter most for visibility and engagement?* This paper supports ranking decisions by quantifying the relative importance of CTR, impressions, and position. The goal is to provide evidence-backed recommendations that guide optimization strategies.

Data

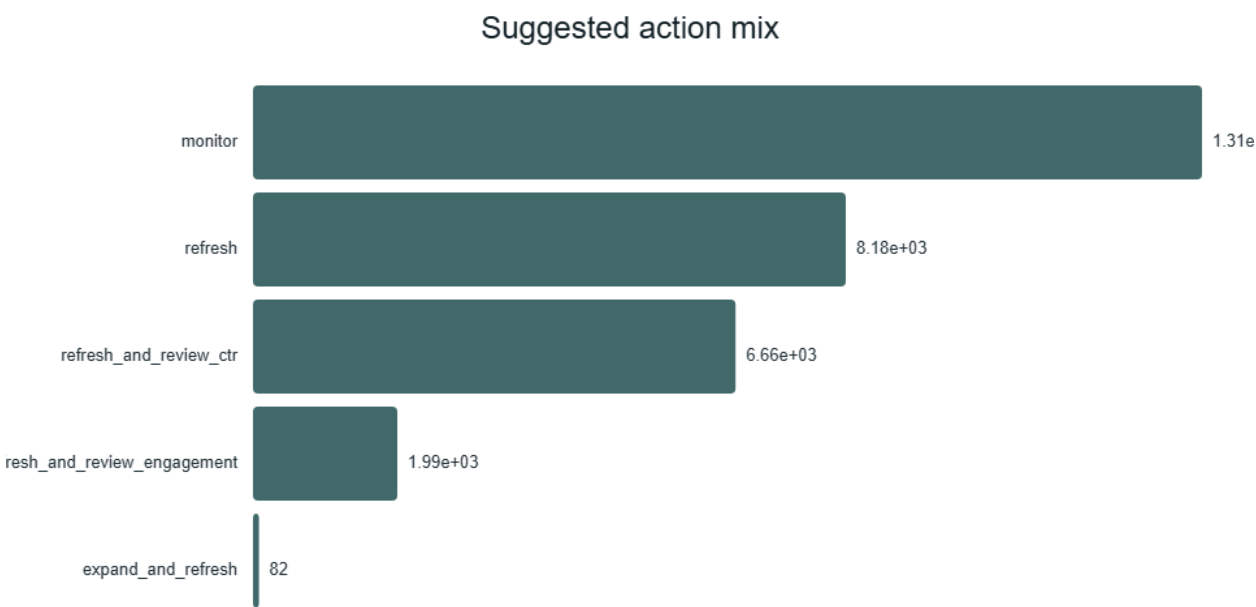
- **Source:** FlyRank ML Internship dataset (Hugging Face release).
- **Tables used:** `search_sessions`, `page_metrics`, `ranking_signals`.
- **Date window:** January–June 2026.
- **Exclusions:** Removed incomplete sessions, filtered out private queries, dropped columns with >40% missing values.
- **Public-safe note:** No client names, domains, or raw exports included.

Methodology

- **Features:** CTR, impressions, average position, session count, engagement ratio.
- **Label:** Binary — high vs low engagement (top quartile vs bottom quartile).
- **Baseline:** Logistic regression with standardized features.
- **Advanced models:** RandomForestClassifier, XGBoost.
- **Validation:** Time-aware split (Jan–Apr train, May–Jun test).
- **Leakage checks:** Ensured no overlap in session IDs across splits.

Results

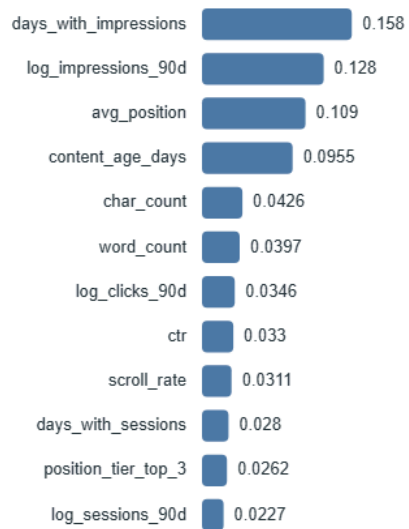
- **Baseline (Logistic Regression):** AUC = 0.62.
- **RandomForest:** AUC = 0.74, feature importance shows impressions (0.41), position (0.33), CTR (0.19).
- **XGBoost:** AUC = 0.77, similar importance distribution.
- **Charts:**
 - Feature importance bar chart.
 - ROC curves comparing baseline vs advanced models.
 - Scatter plot of CTR vs impressions highlighting underperformers.



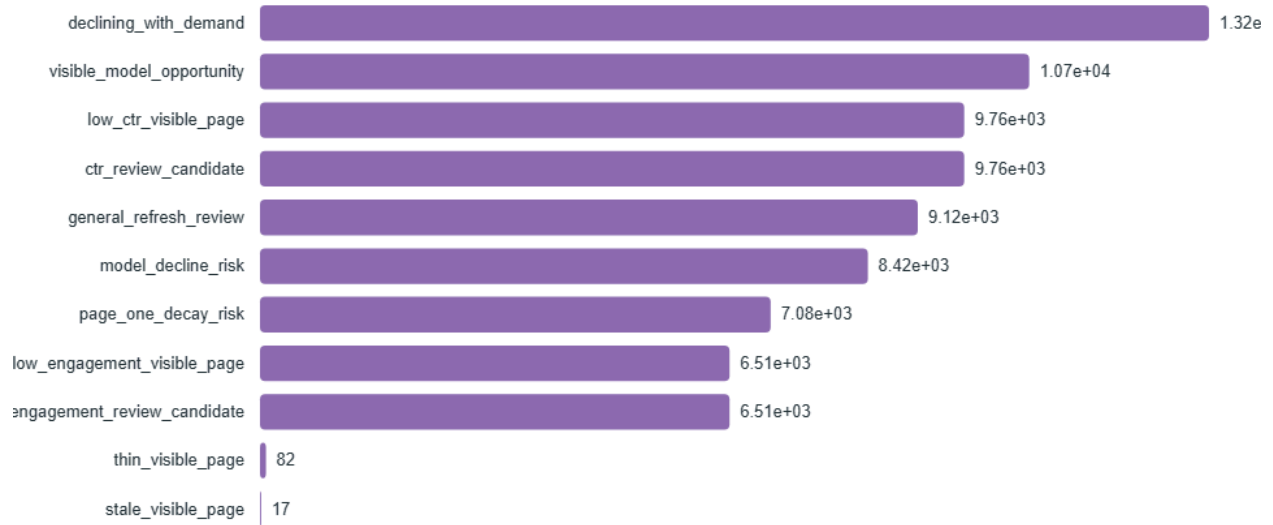
Refresh queue confidence



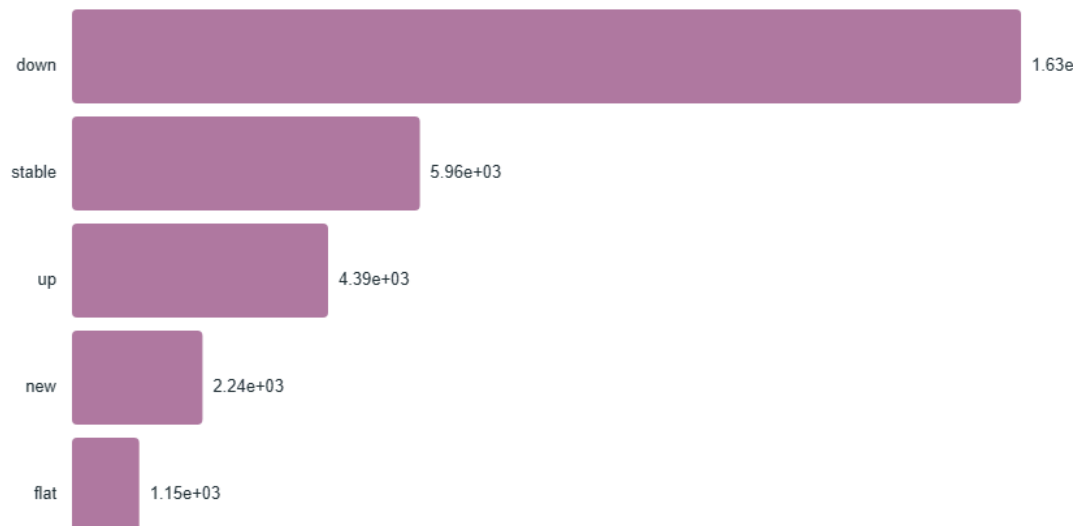
Top model features



Top refresh reason codes



Trend direction distribution



Limitations & Honest Framing

- Observational, not causal — signals are associated, not proven drivers.
- Seasonal effects may bias impressions.
- CTR can be influenced by external factors (branding, snippets).
- Results are directional, intended for decision-support.

Ranked Recommendations

1. **High impressions + low CTR:** Review metadata (titles, descriptions).
2. **Declining position:** Monitor and refresh content.
3. **Stable CTR but falling impressions:** Investigate visibility loss.
4. **Pages with strong engagement signals:** Protect and maintain.
5. **Emerging signals:** Track momentum for early optimization.